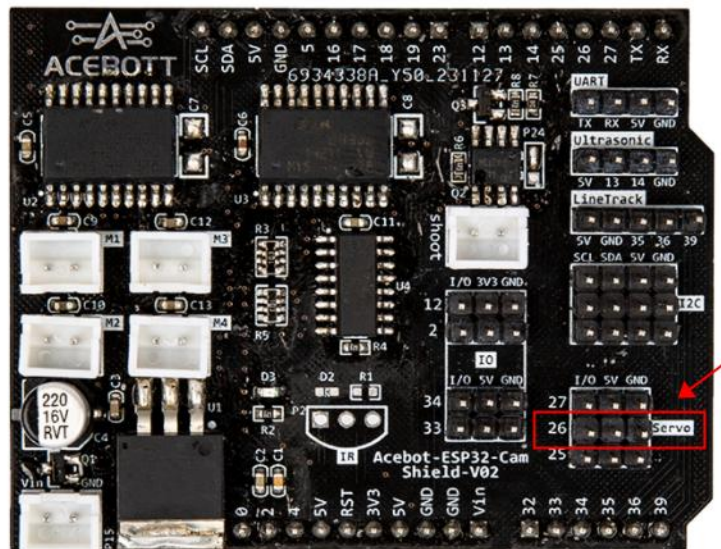


Prepare

1.Servo Angle Calibration Method

Connect the black MG995 servo module to the 26th interface of the ESP32 expansion board, ensuring the gray wire of the servo aligns with GND. Then, use a USB data cable to connect the computer to the ESP32 Controller board. After powering on, upload the servo angle calibration program. This step is to pre-adjust the servo angle to facilitate the subsequent assembly.

[\[Click here to get the servo calibration initialization program\]](#)



2.Assembly Steps

Follow the steps to assemble the water gel blaster car: [\[Click here to get the assembly steps for Shooting Car B\]](#)

Note: If you want to watch the assembly video, please click the link below.

<https://www.youtube.com/playlist?list=PLkW5fEtHNu6JbnSmm2qSQQ3nkMr6SF59Y>

Or scan the QR code below.



3.APP Download (Skip if already downloaded)

(1)If you have an iOS device, search for the keyword "ACEBOTT" in the App Store and download the app. If you have an Android device, search for the keyword "ACEBOTT" in the Google Play Store and download the app. The icon is shown below.

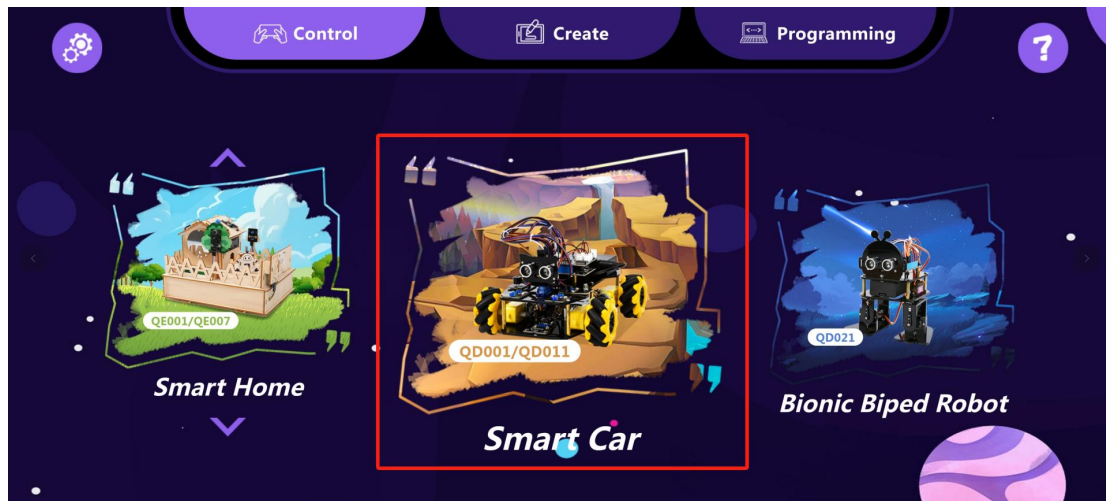


Note: 1. This tutorial is applicable to ACEBOTT APP version 2.0 and above. You can click the settings button in the upper left corner of the APP to view the software version number. Please make sure that the software version you are using meets the requirements; 2. If you need to update the ACEBOTT software version, you can refer to the method prompted in this tutorial to download the latest APP version.

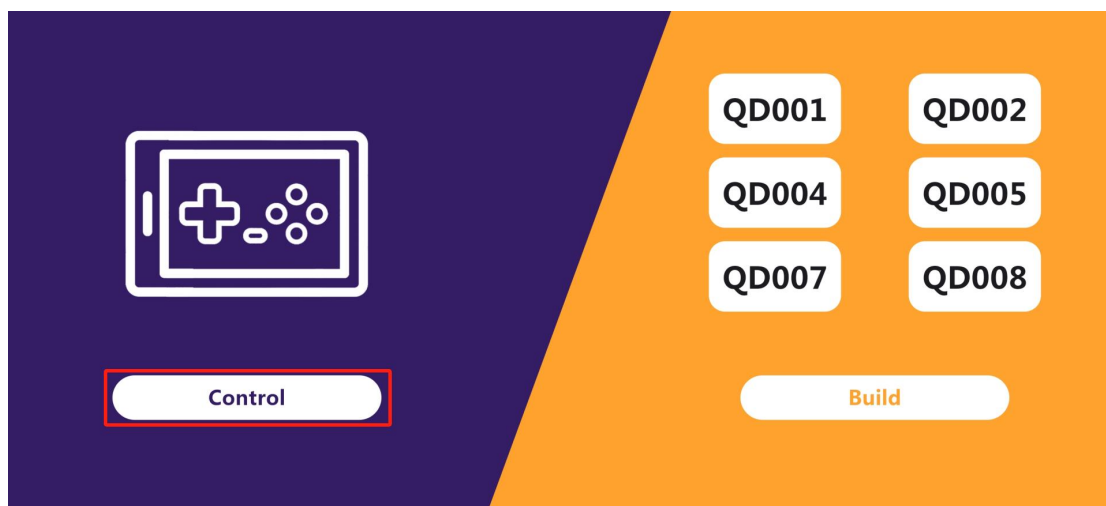
(2)After opening the app, you will enter the splash screen.



(3)Enter the selection interface and select Smart Car.



After clicking to enter, select "Control" to access the car control page.



Note: You can click the build button on the right to view the assembly video of this project.

(4) Enter the water gun car control interface (it cannot be directly controlled yet, and the program needs to be burned).



4.Program burning

(1)Shooting Car APP Control Program

The app cannot control the shooting car directly yet, as the control program needs to be flashed onto the car's ESP32 Controller board first.

The program of the image-transmission shooting car includes the program of the water bomb car itself and the Cam module program, so it needs to correspond to different programs and upload them to different Controller board. Please turn on the power of the car before uploading.

①Smart car program: [\[Click to get the acebott-esp32-car-body program\]](#)

Note: When burning the program to the smart car's Controller board, you need to unplug the TX and RX cables of the camera first, and then connect the TX and RX after the program upload is completed.

②Cam module program: [\[Click to get the acebott-esp32-car-camera program\]](#)

Note:

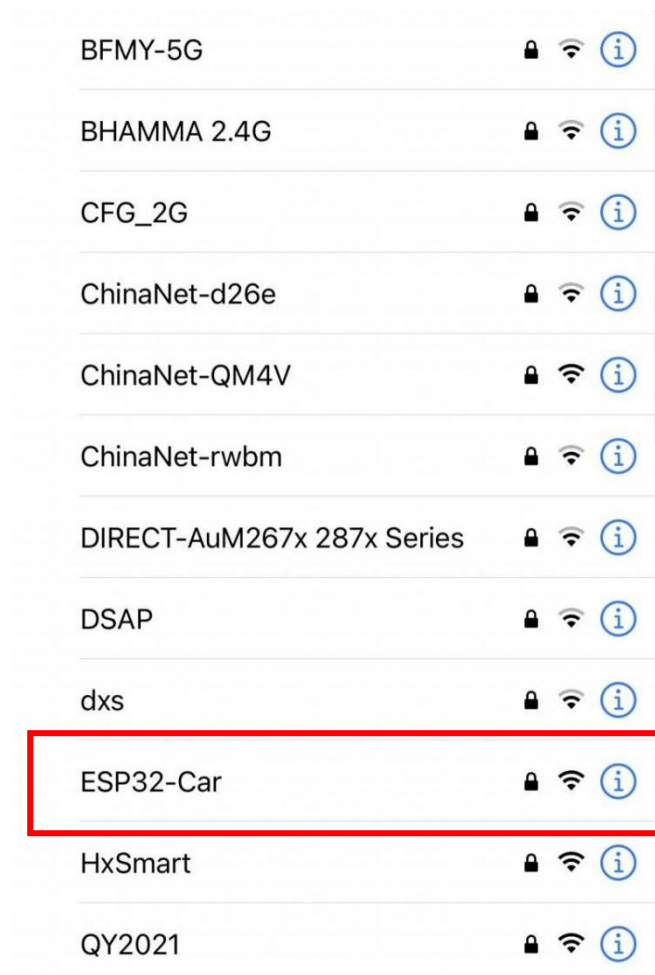
1.Before uploading the program, please confirm whether the firmware of CAM module

has been burned, if not, please click ACEBOTT QD005 ESP32 Shooting Car B\3. Firmware in the "[CAM module install firmware.pdf](#)" file, and follow the prompts to complete the firmware installation.

2.If you need to run offline, you need to run the program on main.py.

(2)Connect to WiFi

Scan for WiFi on your phone by navigating to "Settings" → "WLAN". Connect to the WiFi hotspot named "ESP32-Car" with the password 12345678, as shown below.

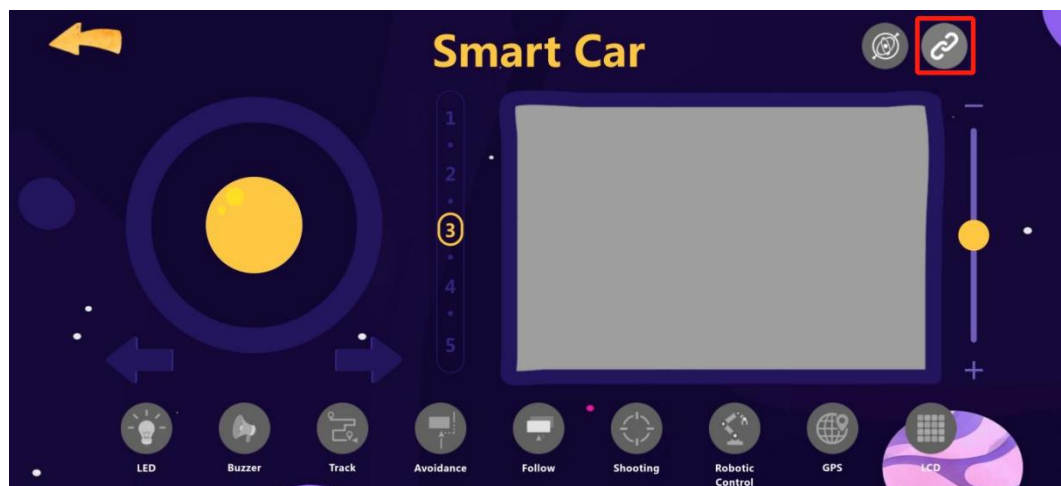


Note:The WiFi name and password can be customized. When we have multiple smart cars, we can distinguish each one by assigning different WiFi names.

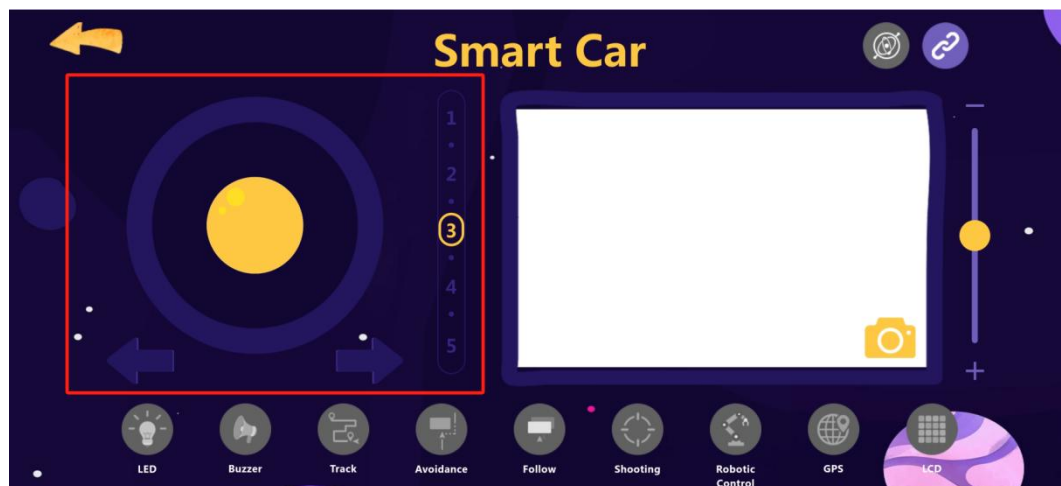
```
ssid = "ESP32-Car"  
password = "12345678"
```

(3)Control with the APP

① After connecting to the WiFi, tap the connection icon in the top right corner of the APP to complete the connection.

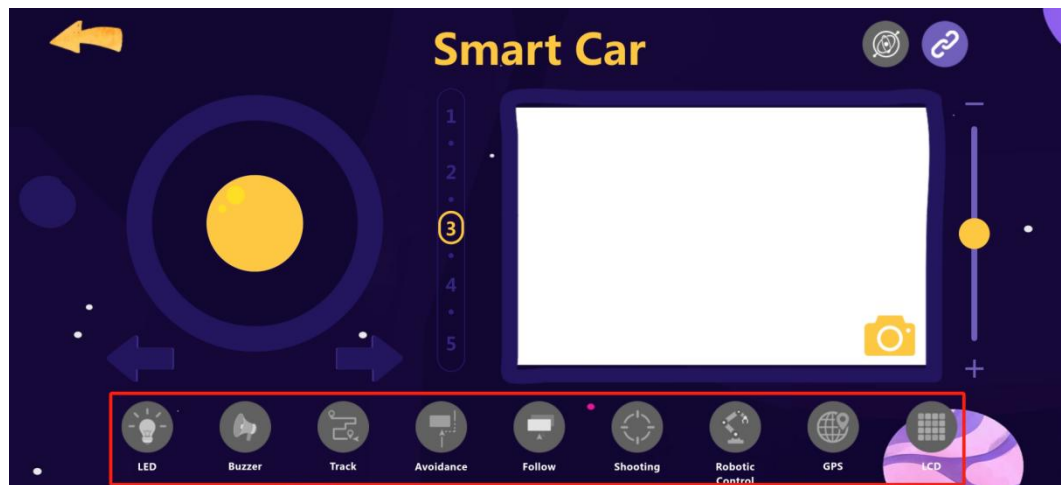


② The left panel controls the basic movements of the car: forward, backward, left move, right move, turn left, turn right; the middle numbers are used for speed adjustment of the car, the larger the number, the faster the speed.



③ Below is a row of buttons for controlling the functions of the car, from left to right, they control the following functions of the car: LED function, music playback (after clicking, there are four types of music to choose from), track mode (after clicking, there are two track modes to choose from), avoidance mode, follow mode, shooting, robotic

arm, GPS, LCD function. Shooting means controlling the firing of a water blaster.

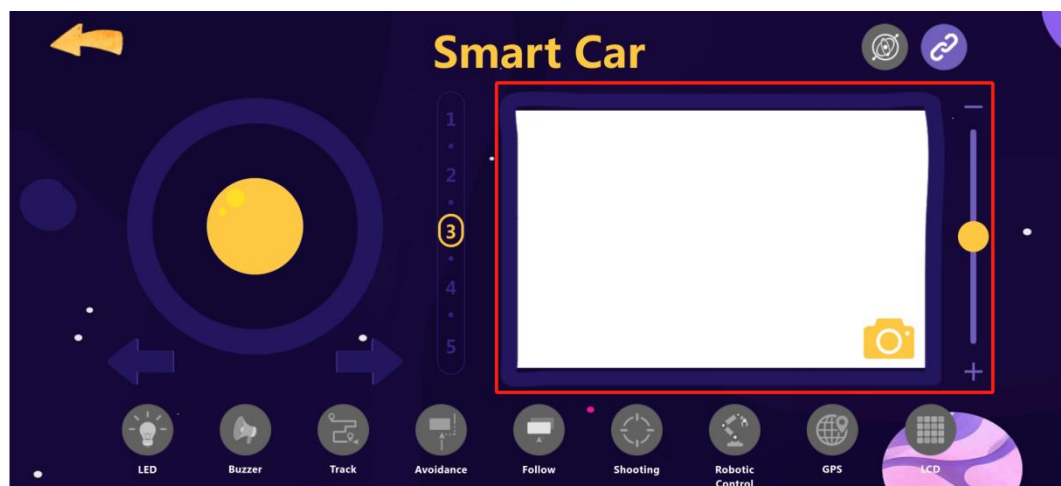


Note:

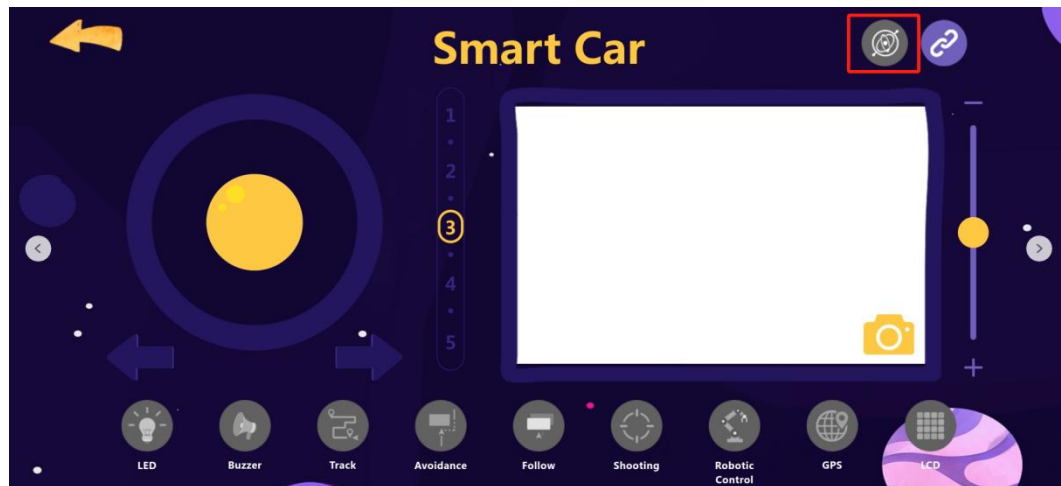
1. After clicking on the track mode, avoidance mode, or Follow mode, the car will enter a continuous tracking, avoidance, or follow state, respectively. To exit this function, you need to click the corresponding function button again to exit the current state. For example, to turn off track mode, you need to click the track button again to exit the track state.

2. Shooting, robotic control, GPS, and LCD are extended functions of the car, which require the corresponding expansion packs to be added for use.

④The central control panel is for the car's camera function, and the slider controls the angle of the gun. Clicking the camera in the bottom right corner allows you to save the screen.



⑤ In the upper right corner of the smart car's operation interface, there is a gyroscope control provided. After clicking this button, you can control the movement of the car through the gyroscope of your mobile phone. If your phone does not have a built-in gyroscope, you can ignore this feature.



5.Other

The first shooting of the water bomb gun may cause no sound due to the internal resistance, do not worry, at this time you can fire several times on the APP.